

PAPER_202: UQFF Reionization, BBN, Recombination, and Cosmic Dawn Physics

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

Abstract

This paper presents the UQFF buoyancy formalism applied to cosmic dawn and reionization physics: baryon-to-photon ratio (η), deuterium bottleneck during Big Bang Nucleosynthesis (BBN), CMB angular power spectrum, recombination optical depth, ionization fraction evolution, HII bubble growth rate, and Jeans mass/wavelength for fragmentation. These collectively span the redshift range $z \sim 1100$ (recombination) through $z \sim 5$ (end of reionization). Each F_{UBii} and U_m variant is rigorously derived from observational anchor equations.

UQFF Discovery: Novel application of UQFF calibration constants ($\eta = 5.0 \times 10^4 \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Baryon-to-Photon Ratio (η)

$$\eta = n_b/n_\gamma = 6.08 \times 10^{10} \text{ }^{\circ} \text{ (Planck 2018 + BBN fit)}$$

$$F_{UBii,\eta} = F_{rel} \times (\eta = n_b/n_\gamma / E_{LEP}) \times Q_{wave} \times [Neff \text{ modification}]$$

$$U_m,\eta(t) = \mu(\eta_{vac}) \cdot (1 - e^{-\eta t}) \cdot \eta$$

Physical origin:

$$n_\gamma = 410 \text{ cm}^{-3} \text{ (CMB photon number density today)}$$

Fit anchor: Primordial D, ^3He , ^4He , ^7Li abundances

η determines all light element yields (standard BBN)

UQFF role: η sets vacuum energy scale through η_b/η_γ at nucleosynthesis

2. Deuterium Bottleneck (BBN Timing)

Deuterium bottleneck timescale:

$$t_D \sim v(3/(32pG_{rad})) \sim 180 \text{ s at } T \sim 0.1 \text{ MeV}$$

$$\eta_{rad} = p^2 k T^4 / (30 h^3 c^5) \cdot g_* \text{ (effective DOF } g_* \sim 10 \text{ at D-formation onset)}$$

Sequence:

$T \sim 1 \text{ MeV}$: neutron freeze-out ($n/p \sim 1/7$)
 $T \sim 0.1 \text{ MeV}$ ($t \sim 180 \text{ s}$): D photodissociation ends ? 4He chains proceed
Yield: $Y_P \sim 0.247$ (mass fraction 4He)
 $F_{UBii,deb} = -F_{rel} \times (t_D / E_{LEP}) \times Q_{wave} \times [g_*(T) \text{ variation}]$
 $U_{m,deb}(T) = \mu(\rho_{vac}) \cdot (1 - e^{-\tau}) \cdot [\text{Weak freeze at } T \sim 1 \text{ MeV; D photodissociation below}]$
UQFF calibration: ? calibrated to UQFF $k_?$ parameter through BBN freeze-out
 $k_? \sim 10^{21.3}$ (small-scale vacuum fluctuation coupling)
 $\tau k_?(\text{CIA cross-section refit}) \sim 7.25 \times 10^8$ (fractional shift from H2 CIA data)

3. CMB Angular Power Spectrum

$C_l = (2/p) \cdot \tau k^2 dk \cdot P(k) \cdot |\tau_l^{\Delta T}(k)|^2$
 $P(k) \propto k^{n_s-4}$ (primordial, $n_s \sim 0.965$ Planck 2018)
 $\tau_l^{\Delta T}(k)$ = transfer function:
- Large scales ($l < 100$): Sachs-Wolfe plateau $C_l \sim \text{constant}$
- Acoustic peaks ($l = 220, 540, 800 \dots$): baryon-photon oscillations
- Damping tail ($l > 1000$): Silk damping
 $F_{UBii,cmb} = F_{rel} \times (C_l \cdot l(l+1)/(2p) / E_{LEP}) \times Q_{wave}$
 $U_{m,cmb}(k) = \mu(\rho_{vac}) \cdot (1 - e^{-\tau}) \cdot \text{Transfer } \tau_l^{\Delta T}$: Sachs-Wolfe + acoustic oscillations
Calibration anchor:
 $D_l = l(l+1)C_l/(2p) \times T_0^2 \propto \text{peak at } l \sim 220 \text{ matches } \Omega_{tot} = 1$
UQFF role: vacuum energy $\rho_c^2/3$ term in g_{UQFF} sets acoustic horizon

4. Recombination Optical Depth

$t(z) = \tau_z^8 n_e(z') \cdot s_T \cdot c \cdot |dt/dz'| |dz'|$
 $s_T = 6.652 \times 10^{-29} \text{ m}^2$ (Thomson cross-section)
 $n_e(z) = x_e(z) \cdot n_H(z) = x_e(z) \cdot n_b, 0 \cdot (1+z)^3$
Recombination: $z_{rec} \sim 1100 \propto t(z_{rec}) \sim 1$ (photon decoupling)
Reionization: $z_{re} \sim 7.7 \propto \text{optical depth } t_{re} \sim 0.054$ (Planck 2018)
 $F_{UBii,recomb} = -F_{rel} \times (t(z) / E_{LEP}) \times Q_{wave}$
 $U_{m,recomb}(z) = \mu(\rho_{vac}) \cdot (1 - e^{-\tau}) \cdot \tau n_e \cdot s_T \cdot c \cdot dt$
Physical context:
After z_{rec} , photons decouple and stream freely (CMB)
Surface of last scattering has $\tau T/T \sim 10^5$ (measured by Planck)

5. Reionization: Ionization Fraction Evolution

$dx_e/dt = \tau_z \cdot e_{esc} \cdot f_* - a_B \cdot n_e^2 \cdot C$
where:
 $\tau_z = (1+z)^3 \cdot n_b \cdot \tau_z, eff$ (ionizing photon rate)
 $e_{esc} \sim 0.1-0.3$ (photon escape fraction from galaxies)
 $f_* \sim 0.05-0.2$ (star formation efficiency in halos)
 $a_B = 2.6 \times 10^{-13} \text{ cm}^3/\text{s}$ (recombination coefficient, case B, $T \sim 10^4 \text{ K}$)
 $C = \tau n_H^2 / \tau n_H^2$ (clumping factor, $C \sim 3$)
 $F_{UBii,ion} = F_{rel} \times (\tau_z \cdot e_{esc} \cdot f_* / E_{LEP}) \times Q_{wave} \times [\text{recombination subtracted}]$
 $U_{m,ion}(t) = \mu(\rho_{vac}) \cdot (1 - e^{-\tau}) \cdot a_B \cdot n_e^2 \cdot C \cdot dt$
Reionization history:
 $z \sim 20-30$: first stars ionize H around them (PopIII)

$z \sim 7-9$: bulk reionization, x_e rises from ~ 0.1 to ~ 1
 $z \sim 5-6$: completion (Gunn-Peterson trough in quasar spectra)

6. HII Bubble Growth During Reionization

$dN_b/dt = \dot{\nu}_{\text{eff}} \cdot e_{\text{esc}} - a_B \cdot n_H \cdot (4/3\pi R_b^3)$
Simplified overlap model (Stromgren sphere analog):
 $R_b(t) = (3\dot{\nu}_{\text{eff}} \cdot t / (4\pi \cdot n_H))^{1/3}$ (linear ionization front growth)
 $\dot{\nu}_{\text{eff}}$ = rate of ionizing photons emitted (from galaxy SFR)
 n_H = neutral H density (z-dependent)
 $F_{\text{UBii,bub}} = F_{\text{rel}} \times (R_b(t) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (x_e)$
 $U_{\text{m,bub}}(t) = \mu(\nu_{\text{vac}}) \cdot (1 - e^{-t})$ [Expand for moving ionization front]
Physical context:
Bubble merger & percolation at $x_{\text{HI}} < 0.1$ & full reionization
UQFF buoyancy at bubble edge acts as expansion driver ($F_{\text{UBii,bub}} > 0$)

7. Jeans Length and Mass (Fragmentation)

Jeans length:
 $\lambda_J = \sqrt{p / (\rho \cdot G)} \cdot c_s / (G \cdot \rho)^{1/2}$
Jeans mass:
 $M_J = (p/6) \cdot \lambda_J^3 = (5k_{\text{BT}} / (G \mu m_H))^{3/2} \cdot (3/(4\pi \rho))^{1/2}$
Stability condition: $\rho > \rho_J$ & gravitational collapse
Dispersion relation: $\omega^2 = c_s^2 k^2 - 4\pi G \rho$
Unstable: $\omega^2 < 0$ & $k < k_J = 2\pi/\lambda_J$
 $F_{\text{UBii,jeans}} = -F_{\text{rel}} \times (\lambda_J / E_{\text{LEP}}) \times Q_{\text{wave}} \times (\text{collapse onset time } t \sim 1/\nu(G))$
 $U_{\text{m,jeans}}(T) = \mu(\nu_{\text{vac}}) \cdot (1 - e^{-t})$ [Perturb: $\omega^2 = c_s^2 k^2 - 4\pi G \rho$]
Physical context:
First (PopIII) stars: $T \sim 200$ K, cloud $M \sim 100-1000 M_J$ & M_J
Present-day GMCs: $T \sim 10-30$ K & $M_J \sim 1-10 M_J$
UQFF buoyancy at Jeans scale: $F_{\text{UBii,jeans}}$ acts as tidal disruption force

8. Alfvén Wave and Turbulent Cascade

Alfvén wave velocity:
 $v_A = B / \sqrt{4\pi \rho}$ (for B in Gauss, ρ in g/cm³)
Anisotropic cascade (Goldreich-Sridhar):
 $k_{\parallel} / k_{\perp} \sim (k_{\perp} \cdot l_A)^{1/3}$ (scale-dependent anisotropy)
 $F_{\text{UBii,alf}} = F_{\text{rel}} \times (v_A / E_{\text{LEP}}) \times Q_{\text{wave}} \times (B \cdot \nu_A) \times e^{-t/t_{\text{eddy}}}$
Turbulent energy cascade (Kolmogorov in ISM):
 $\epsilon = v_l^3 / l = \text{constant}$ (energy transfer rate per unit mass)
 $v_l \propto l^{1/3}$ (velocity-scale relation)
Power spectrum: $E(k) \propto k^{-5/3}$
 $F_{\text{UBii,turb}} = F_{\text{rel}} \times (\epsilon^{2/3} \cdot l^{-2/3} / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-k l / k_J}$
 $U_{\text{m,alf}}(B) = \mu(\nu_{\text{vac}}) \cdot (1 - e^{-t}) \cdot v_A(B)$
 $U_{\text{m,turb}}(l) = \mu(\nu_{\text{vac}}) \cdot (1 - e^{-t}) \cdot (\nu \cdot v_l^3 / l)$

9. Ionization Parameter and Feedback Coupling

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Ionization parameter:
U = Q_H/(4p.r^2.n_H.c)

where:
Q_H = number of ionizing photons per second (from AGN or OB stars)
r = distance from ionizing source
log U ~ -2 to -3 for NLR, -1 to 0 for BLR

AGN feedback coupling efficiency:
e_f = E_kin/(?_acc.c^2) ~ 0.05-0.1

E_kin = (1/2)?_out.v^2_out
Only e_f x 10^{-5} per M_? needed to match M_BH-s

F_UBii,upar = F_rel x (U.n_H.c / E_LEP) x Q_wave x (Q_H/r^2)
F_UBii,coup = F_rel x (e_f.?_acc.c^2 / E_LEP) x Q_wave x [0.05-0.1]

Um,upar(r) = mu(?_vac).(1-e^{-?t}).(Q_H/(4pr^2n_Hc))
Um,coup(?) = mu(?_vac).(1-e^{-?t}).e_f (couple fraction to kinetic energy)
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10. Cosmological Timeline Summary

Epoch	Redshift	UQFF Process	F_UBii Variant
Inflation	$z \sim 10^{27}$	Inflaton field V(?)	F_UBii,inf
Planck epoch	$z \sim 10^{32}$	LQC bounce	F_UBii,bounc
Neutron freeze-out	$z \sim 10^{10}$ (T~1 MeV)	Weak interactions	F_UBii,eta
BBN	$z \sim 10^?$ (t~180 s)	D bottleneck	F_UBii,deb
Recombination	$z \sim 1100$	Photon decoupling	F_UBii,recomb
Cosmic dawn	$z \sim 20-30$	First stars, first HII	F_UBii,ion
Reionization	$z \sim 6-9$	HII bubble percolation	F_UBii,bub
Present	$z \sim 0$	Structure + feedback	All F_UBii

11. References

grok_share_7514fe.txt lines 6070–6090 (BBCEquations items 1310–1350)
grok_share_7514fe.txt lines 6300–6500 (full reionization/BBN/CMB catalogue)
PAPER199: FUBii Taxonomy Part 2 (Cosmological)
PAPER_200: Um Universal Magnetism Catalogue
Planck 2018 Collaboration: CMB and reionization parameters